

# TRIGONOMETRIC RATIOS



## SYNOPSIS

- **Angle:** An angle is the union of two rays having a common end point in a plane.
- **Measurement of an angle :**
- **Sexagesimal system:**
  - (i) One right angle =  $\frac{\pi}{2}$  radian =  $90^\circ$ .
  - (ii)  $\pi$  radian = 2 right angles =  $180^\circ$ .
  - (iii)  $1^\circ = 60$  minutes ( $60'$ )
  - (iv)  $1' = 60$  seconds ( $60''$ )
  - (v)  $1^\circ = 0.001745$  radian
  - (vi)  $1$  radian  $\cong 57^\circ 17' 45''$  (approx)
- **Centesimal system:**
  - (i) 1 right angle = 100 grades written as  $100^g$
  - (ii) 1 grade or  $1^g = 100$  minutes ( $100'$ )
  - (iii) 1 minute or  $1' = 100$  seconds ( $100''$ )
- **Circular system :**
- **Radian:** A radian is the angle subtended at the centre of a circle by an arc equal in length to the radius of the circle. The length of arc  $l = r\theta$ .
  - (i) 1 revolution =  $2\pi$  radians =  $360^\circ$
  - (ii)  $\pi$  radians = 2 right angles =  $2 \times 90^\circ = 180^\circ$
  - (iii) 1 degree ( $1^\circ$ ) =  $\frac{\pi}{180}$  rad  $\cong 0.01745$  rad
  - (iv) 1 rad ( $1^c$ ) =  $\frac{180}{\pi}$  degrees  $\cong 57^\circ 17' 46''$
- **Note:** (i) Value of  $\pi = \frac{22}{7}$  (or)  $\frac{355}{113}$  (or) 3.1416
  - (ii)  $\pi$  is an irrational number
  - (iii)  $\pi = \frac{\text{circumference of the circle}}{\text{diameter of the circle}}$
- **Trigonometric Identities :**
  - i)  $(\sin \theta)(\operatorname{cosec} \theta) = 1, \theta \neq n\pi, n \in \mathbb{Z}$

$$\text{ii) } (\cos \theta)(\sec \theta) = 1, \theta \neq (2n+1)\frac{\pi}{2}, n \in \mathbb{Z}$$

$$\text{iii) } (\tan \theta)(\cot \theta) = 1, \theta \neq (2n+1)\frac{\pi}{2}, \theta \neq n\pi$$

$$\text{iv) } \sin^2 \theta + \cos^2 \theta = 1 \Rightarrow \sin^2 \theta = 1 - \cos^2 \theta$$

$$\text{v) } \sec^2 \theta - \tan^2 \theta = 1 \Rightarrow \sec^2 \theta = 1 + \tan^2 \theta$$

$$\left( \theta \neq (2n+1)\frac{\pi}{2}, n \in \mathbb{Z} \right).$$

$$\text{(vi) } \operatorname{cosec}^2 \theta - \cot^2 \theta = 1$$

$$\Rightarrow \operatorname{cosec}^2 \theta = 1 + \cot^2 \theta \quad (\theta \neq n\pi, n \in \mathbb{Z})$$

- **Note:** (i) If  $\theta \neq (2n+1)\frac{\pi}{2}, n \in \mathbb{Z}$ , then

$$\sec^2 \theta - \tan^2 \theta = 1 \Rightarrow (\sec \theta + \tan \theta)(\sec \theta - \tan \theta) = 1$$

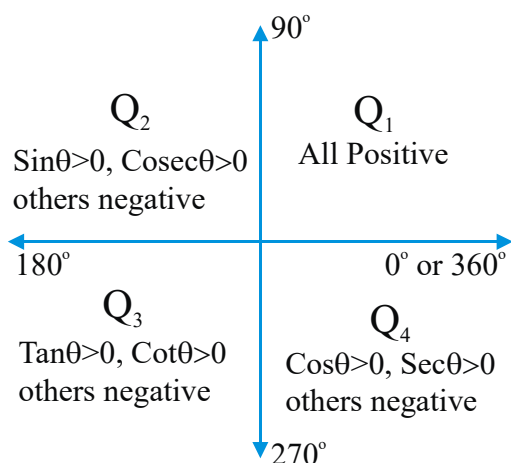
$$\Rightarrow \sec \theta + \tan \theta = \frac{1}{\sec \theta - \tan \theta}$$

- (ii) If  $\theta \neq n\pi, n \in \mathbb{Z}$  then

$$\operatorname{cosec} \theta - \cot \theta = \frac{1}{\operatorname{cosec} \theta + \cot \theta}$$

## Trigonometric ratios of various angles:

| Trig. Ratio                   | $0^\circ$ | $30^\circ$           | $45^\circ$           | $60^\circ$           | $90^\circ$ |
|-------------------------------|-----------|----------------------|----------------------|----------------------|------------|
| $\sin \theta$                 | 0         | $\frac{1}{2}$        | $\frac{1}{\sqrt{2}}$ | $\frac{\sqrt{3}}{2}$ | 1          |
| $\cos \theta$                 | 1         | $\frac{\sqrt{3}}{2}$ | $\frac{1}{\sqrt{2}}$ | $\frac{1}{2}$        | 0          |
| $\tan \theta$                 | 0         | $\frac{1}{\sqrt{3}}$ | 1                    | $\sqrt{3}$           | $\infty$   |
| $\operatorname{cosec} \theta$ | $\infty$  | 2                    | $\sqrt{2}$           | $\frac{2}{\sqrt{3}}$ | 1          |
| $\sec \theta$                 | 1         | $\frac{2}{\sqrt{3}}$ | $\sqrt{2}$           | 2                    | $\infty$   |
| $\cot \theta$                 | $\infty$  | $\sqrt{3}$           | 1                    | $\frac{1}{\sqrt{3}}$ | 0          |



**Note :** i)  $\sin n\pi = \tan n\pi = \cos(2n+1)\frac{\pi}{2} = 0, \forall n \in \mathbb{Z}$

ii)  $\sin(2n+1)\frac{\pi}{2} = (-1)^n$  and  $\cos n\pi = (-1)^n, \forall n \in \mathbb{Z}$

➤ **Domain and range of trigonometric functions :**

| Trig. function | Domain                                                              | Range                            |
|----------------|---------------------------------------------------------------------|----------------------------------|
| sin            | $\mathbb{R}$                                                        | $[-1, 1]$                        |
| cos            | $\mathbb{R}$                                                        | $[-1, 1]$                        |
| tan            | $\mathbb{R} - \left\{(2n+1)\frac{\pi}{2}, n \in \mathbb{Z}\right\}$ | $\mathbb{R}$                     |
| cot            | $\mathbb{R} - \{n\pi, n \in \mathbb{Z}\}$                           | $\mathbb{R}$                     |
| cosec          | $\mathbb{R} - \{n\pi, n \in \mathbb{Z}\}$                           | $(-\infty, -1] \cup [1, \infty)$ |
| sec            | $\mathbb{R} - \left\{(2n+1)\frac{\pi}{2}, n \in \mathbb{Z}\right\}$ | $(-\infty, -1] \cup [1, \infty)$ |

**W.E-1:** If  $\sin \theta = \frac{a-4}{2}$  then  $a$  lies in

**Sol:**  $-1 \leq \sin \theta \leq 1 \Rightarrow -1 \leq \frac{a-4}{2} \leq 1$

$\Rightarrow -2 \leq a-4 \leq 2 \Rightarrow 2 \leq a \leq 6 \Rightarrow a \in [2, 6]$

➤ **Some useful results :**

(a) If  $A+B=90^\circ$  or  $270^\circ$ , then

(i)  $\sin^2 A + \sin^2 B = 1$  (ii)  $\cos^2 A + \cos^2 B = 1$

(iii)  $\tan A \cdot \tan B = 1$  (iv)  $\cot A \cdot \cot B = 1$

(b) If  $A+B=180^\circ$ , then

(i)  $\cos A + \cos B = 0$  (ii)  $\sin A - \sin B = 0$

(iii)  $\tan A + \tan B = 0$

(c) If  $A+B=360^\circ$ , then

(i)  $\sin A + \sin B = 0$  (ii)  $\cos A - \cos B = 0$

(iii)  $\tan A + \tan B = 0$

**W.E-2:**  $\tan 130^\circ \cdot \tan 140^\circ =$

**Sol:**  $130^\circ + 140^\circ = 270^\circ$

$\therefore \tan 130^\circ \cdot \tan 140^\circ = 1$

**W.E-3:**  $\sin^2 55^\circ + \sin^2 35^\circ =$

**Sol:**  $55^\circ + 35^\circ = 90^\circ$

$\therefore \sin^2 55^\circ + \sin^2 35^\circ = 1$

➤ (i) If  $a \cos \theta + b \sin \theta = c$  and

$a \sin \theta - b \cos \theta = k$  then  $a^2 + b^2 = c^2 + k^2$

(ii) If  $a \sec \theta + b \tan \theta = c$  and

$a \tan \theta + b \sec \theta = k$  then  $a^2 - b^2 = c^2 - k^2$

(iii) If  $a \operatorname{cosec} \theta + b \cot \theta = c$  and

$a \cot \theta + b \operatorname{cosec} \theta = k$  then  $a^2 - b^2 = c^2 - k^2$

**W.E-4:** If  $8 \cos \theta + 6 \sin \theta = 5$  then  $8 \sin \theta - 6 \cos \theta =$

**Sol:** Let  $8 \sin \theta - 6 \cos \theta = k$

$a^2 + b^2 = c^2 + k^2$

$\therefore 8^2 + 6^2 = 5^2 + k^2$

$\Rightarrow k^2 = 75 \Rightarrow k = \pm 5\sqrt{3}$

➤  $\sqrt{x^2} = |x| = \begin{cases} x & \text{if } x > 0 \\ -x & \text{if } x < 0 \\ 0 & \text{if } x = 0 \end{cases}$

for example the value of

$\sqrt{\cos^2 100^\circ} = |\cos 100^\circ|$

$= -\cos 100^\circ (\because \cos 100^\circ < 0)$

➤ (i)  $\sin \theta + \sin(\pi + \theta) + \sin(2\pi + \theta) + \dots$

$\dots + \sin(n\pi + \theta) = \begin{cases} 0 & \text{if } n \text{ is odd} \\ \sin \theta & \text{if } n \text{ is even} \end{cases}$

(ii)  $\cos \theta + \cos(\pi + \theta) + \cos(2\pi + \theta) + \dots$

$\dots + \cos(n\pi + \theta) = \begin{cases} 0 & \text{if } n \text{ is odd} \\ \cos \theta & \text{if } n \text{ is even} \end{cases}$

**W.E-5:**  $\sin \theta + \sin(\pi + \theta) + \dots + \sin(100\pi + \theta) =$

**Sol:**  $n = 100$  is even

$\therefore$  the required value  $= \sin \theta$

**W.E-6:**  $\cos \theta + \cos(\pi + \theta) + \dots + \cos(299\pi + \theta) =$

**Sol:**  $n = 299$  is odd

$\therefore$  required value  $= 0$

➤ If  $a > 0, b > 0$  and  $f(x) > 0$  then

$$af(x) + \frac{b}{f(x)} \geq 2\sqrt{ab}$$

**Proof:** We know that A.M.  $\geq$  G.M.

$$\Rightarrow \frac{af(x) + \frac{b}{f(x)}}{2} \geq \sqrt{af(x) \cdot \frac{b}{f(x)}}$$

$$\Rightarrow af(x) + \frac{b}{f(x)} \geq 2\sqrt{ab}$$

➤ **Note:** (i)  $x + \frac{1}{x} \geq 2$ , for  $x > 0$

(ii)  $a \tan x + b \cot x \geq 2\sqrt{ab}$

(iii)  $25 \cos x + 16 \sec x \geq 2\sqrt{25 \times 16} = 40$

➤ (i)  $\cot^2 \theta - \cos^2 \theta = \cot^2 \theta \cos^2 \theta$

(ii)  $\tan^2 \theta - \sin^2 \theta = \tan^2 \theta \sin^2 \theta$

(iii)  $\cos^2 \theta + \sec^2 \theta = \cos^2 \theta \cdot \sec^2 \theta$

➤ (i)  $\sin^4 \theta + \cos^4 \theta = 1 - 2 \sin^2 \theta \cos^2 \theta$

(ii)  $\sin^6 \theta + \cos^6 \theta = 1 - 3 \sin^2 \theta \cos^2 \theta$

(iii)  $\sin^2 \theta + \cos^4 \theta = 1 - \sin^2 \theta \cos^2 \theta$

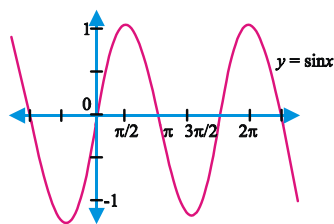
(iv)  $\cos^2 \theta + \sin^4 \theta = 1 - \sin^2 \theta \cos^2 \theta$

**W.E-7:**  $\frac{2 + \sin^6 \theta + \cos^6 \theta}{\sin^2 \theta + \cos^4 \theta} =$

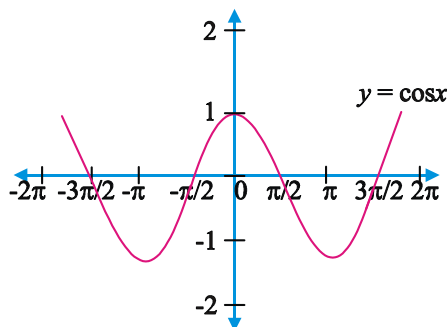
**Sol:** 
$$\frac{2 + (1 - 3 \sin^2 \theta \cos^2 \theta)}{1 - \sin^2 \theta \cos^2 \theta}$$
  

$$= \frac{3(1 - \sin^2 \theta \cos^2 \theta)}{(1 - \sin^2 \theta \cos^2 \theta)} = 3$$

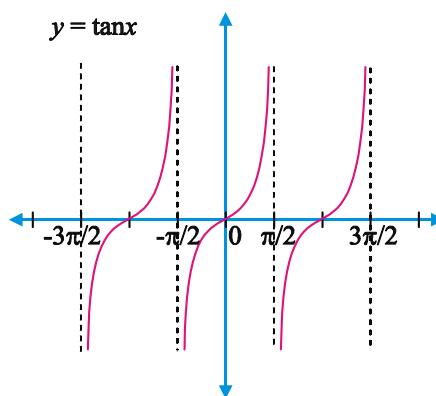
➤ **Graph of  $\sin x$ :**



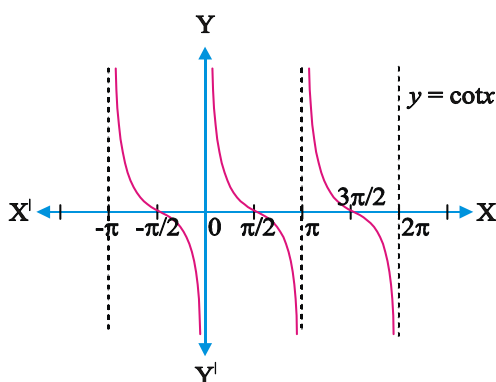
➤ **Graph of  $\cos x$ :**



➤ **Graph of  $\tan x$ :**



➤ **Graph of  $\cot x$ :**





LEVEL - I (C.W)

- $-13\pi/6$  radians =  
1)  $-390^\circ$  2)  $-490^\circ$  3)  $-410^\circ$  4)  $-30^\circ$
- $\cot(1358^\circ) + \tan(3608^\circ) =$   
1) -1 2) 0 3) 1 4) 2
- $\frac{\sin(-660^\circ) \tan(1050^\circ) \sec(-420^\circ)}{\cos(225^\circ) \operatorname{cosec}(315^\circ) \cos(510^\circ)} =$   
1)  $\frac{\sqrt{3}}{4}$  2)  $\frac{\sqrt{3}}{2}$  3)  $\frac{2}{\sqrt{3}}$  4)  $\frac{4}{\sqrt{3}}$
- $\sin 48^\circ \sec 42^\circ + \cos 48^\circ \operatorname{cosec} 42^\circ =$   
1) 0 2) 2 3) 1 4) -1
- $\log \tan 17^\circ + \log \tan 37^\circ + \log \tan 53^\circ + \log \tan 73^\circ =$   
1) 0 2) 1 3) 2 4) 3
- $\cos 24^\circ + \cos 55^\circ + \cos 125^\circ + \cos 204^\circ =$   
1) -1 2) 0 3) 1 4) 2
- $\sec A + \tan A = 3 \Rightarrow \sec A =$   
1)  $\frac{10}{3}$  2)  $\frac{5}{3}$  3)  $\frac{2}{3}$  4)  $\frac{4}{3}$
- $\sec \theta - \tan \theta = 3 \Rightarrow \theta$  lies in the quadrant  
1) I 2) II 3) III 4) IV
- $\sin 10^\circ + \sin 20^\circ + \sin 30^\circ + \dots + \sin 360^\circ =$   
1) 0 2) 1 3) -1 4) 2
- $3[\sin x - \cos x]^4 + 6[\sin x + \cos x]^2 + 4[\sin^6 x + \cos^6 x] =$   
1) 3 2) 6 3) 4 4) 13
- $\frac{\sin^2 \alpha}{1 + \cot^2 \alpha} + \frac{\tan^2 \alpha}{(1 + \tan^2 \alpha)^2} + \cos^2 \alpha =$   
1) -1 2) 0 3) 1 4) 2
- $1 - \frac{\sin^2 x}{1 + \cos x} + \frac{1 + \cos x}{\sin x} - \frac{\sin x}{1 - \cos x} =$   
1)  $\sin x$  2)  $\cos x$   
3)  $\operatorname{cosec} x$  4)  $\sec x$
- If A, B, C are the angles of a triangle ABC  
then  $\cos\left(\frac{3A + 2B + C}{2}\right) + \cos\left(\frac{A - C}{2}\right) =$   
1) 0 2) 1 3)  $\cos A$  4)  $\cos C$

- $\frac{\cot \theta + \operatorname{cosec} \theta - 1}{\cot \theta - \operatorname{cosec} \theta + 1} =$   
1)  $\frac{\sin \theta}{1 + \cos \theta}$  2)  $\frac{\sin \theta}{1 - \cos \theta}$   
3)  $\frac{1 - \cos \theta}{\sin \theta}$  4)  $\frac{\cos \theta}{1 - \sin \theta}$
- $\sin^2(51^\circ - x) + \sin^2(39^\circ + x) =$   
1) -1 2) 0 3) 1 4) 2
- $\frac{\sin^2 \frac{\pi}{18} + \sin^2 \frac{\pi}{9} + \sin^2 \frac{7\pi}{18} + \sin^2 \frac{4\pi}{9}}{\cos^2 \frac{\pi}{18} + \cos^2 \frac{\pi}{9} + \cos^2 \frac{7\pi}{18} + \cos^2 \frac{4\pi}{9}} =$   
1) 1 2) 2 3) 3 4) 4
- If ABCD is a cyclic quadrilateral, then  
 $\cos(180^\circ + A) + \cos(180^\circ - B) + \cos(180^\circ - C) - \sin(90^\circ - D) =$   
1) -1 2) 0 3) 1 4) 2
- $\left[ \sin\left(\frac{\pi}{2} - x\right) + \sin(\pi - x) \right]^2 + \left[ \cos\left(\frac{3\pi}{2} - x\right) + \cos(2\pi - x) \right]^2 =$   
1) 0 2) 2 3) 4 4) 8
- $\sin \theta_1 + \sin \theta_2 + \sin \theta_3 = 3 \Rightarrow$   
 $\cos \theta_1 + \cos \theta_2 + \cos \theta_3 =$   
1) 0 2) 1 3) 2 4) 3
- $8 \sin^2 x + 3 \cos^2 x = 5 \Rightarrow \cot x =$   
1)  $\pm \frac{1}{\sqrt{2}}$  2)  $\pm \frac{1}{\sqrt{3}}$  3)  $\pm \sqrt{\frac{3}{2}}$  4)  $\pm \sqrt{\frac{2}{3}}$
- $\sin x + \sin^2 x = 1 \Rightarrow \cos^2 x + \cos^4 x =$   
1) 0 2) 1 3) 2 4) -1
- If  $\cos(x - y) = -1$ , then  $\cos x + \cos y =$   
1) 0 2) 1 3) 2 4) -1
- If  $\sin x + \operatorname{cosec} x = 2$  then  $\sin^8 x + \operatorname{cosec}^8 x =$   
1) 1 2) 2 3) 3 4) 4
- If  $\sin x + \cos x = a$  then  $|\sin x - \cos x| =$   
1)  $\sqrt{1 - a^2}$  2)  $\sqrt{a^2 - 1}$  3)  $\sqrt{2 - a^2}$  4)  $\sqrt{a^2 - 2}$

25.  $(1 + \sin A)(1 + \sin B)(1 + \sin C) = (1 - \sin A)$

$(1 - \sin B)(1 - \sin C) = k \Rightarrow k =$

1)  $+\sin A \sin B \sin C$

2)  $\pm \cos A \cos B \cos C$

3)  $\pm \sec A \sec B \sec C$

4)  $\pm \operatorname{cosec} A \operatorname{cosec} B \operatorname{cosec} C$

26.  $\operatorname{cosec}^2 A \cot^2 A - \sec^2 A \tan^2 A -$

$(\cot^2 A - \tan^2 A)(\sec^2 A \operatorname{cosec}^2 A - 1) =$

1) 0      2) 1      3) -1      4) 2

27.  $\tan^2 \theta + \sec \theta = 5 \Rightarrow \sec \theta =$

1) 3      2) 2      3) 1      4) -1

28.  $\log \sin 1^\circ \cdot \log \sin 2^\circ \dots \log \sin 179^\circ$

1) 1      2) 0      3) -1      4) 2

29. If  $\tan(\alpha + \beta) = \sqrt{3}$ ,  $\tan(\alpha - \beta) = 1$  then

$\tan 6\beta =$

1) -1      2) 0      3) 1      4) 2

30. If  $x = r \cos \theta \cos \phi$ ,  $y = r \cos \theta \sin \phi$ ,

$z = r \sin \theta$  then  $x^2 + y^2 + z^2 =$

1)  $r^2$       2)  $y^2$       3)  $x^2$       4)  $z^2$

31.  $\sin^2 A \cos^2 B + \cos^2 A \sin^2 B$

$+ \sin^2 A \sin^2 B + \cos^2 A \cos^2 B =$

1) -1      2) 0      3) 1      4) 2

32. If  $2 \sin^2 A - \cot^2 B = 1$  then  $\sin A \sin B =$

1)  $-\frac{1}{\sqrt{2}}$       2)  $\pm \frac{1}{\sqrt{2}}$       3)  $\frac{1}{\sqrt{2}}$       4)  $\frac{-1}{2\sqrt{2}}$

33. The value of

$\sin^6 \theta + \cos^6 \theta + 3 \sin^2 \theta \cos^2 \theta$  is

1) 0      2) 1      3) 2      4) 3

34. If  $x \tan^2 120^\circ + 4 \cos^2 150^\circ = 9$  then  $x =$

1) 3      2) 1      3) 2      4) 4

35. In  $\triangle ABC$ , right angled at C, then

$\tan A + \tan B =$

1)  $\frac{a^2}{bc}$       2)  $\frac{b^2}{ac}$       3)  $\frac{c^2}{ab}$       4)  $\frac{ab}{c}$

36. If  $a = \sin 170^\circ + \cos 170^\circ$  then

1)  $a > 0$       2)  $a < 0$       3)  $a = 0$       4)  $a = 1$

37.  $\tan \frac{\pi}{24} \tan \frac{3\pi}{24} \tan \frac{5\pi}{24} \tan \frac{7\pi}{24} \tan \frac{9\pi}{24} \tan \frac{11\pi}{24} \tan \frac{13\pi}{24} \tan \frac{15\pi}{24} =$

1) 1      2) -1      3)  $\sqrt{3}$       4)  $-\sqrt{3}$

38.  $x = a \sec^3 \theta \tan \theta$ ,  $y = b \tan^3 \theta \sec \theta$

then  $\sin^2 \theta =$

1)  $\frac{x}{a} - \frac{y}{b}$       2)  $\frac{x}{a} + \frac{y}{b}$       3)  $\frac{xy}{ab}$       4)  $\frac{ay}{bx}$

39. If  $\frac{x}{a} \cos \theta + \frac{y}{b} \sin \theta = 1$ ,  $\frac{x}{a} \sin \theta - \frac{y}{b} \cos \theta = 1$

then  $\frac{x^2}{a^2} + \frac{y^2}{b^2} =$

1) 1      2) -1      3) 2      4) 3

40. In a  $\triangle ABC$ , if  $\cot A \cot B \cot C > 0$  then the triangle is

1) acute angled      2) right angled  
3) obtuse angled      4) can't be decided

41. If  $\alpha, \beta$  are complementary angles and

$\sin \alpha = \frac{3}{5}$ , then  $\sin \alpha \cos \beta - \cos \alpha \sin \beta =$

1)  $\frac{7}{25}$       2)  $-\frac{7}{25}$       3)  $\frac{25}{7}$       4)  $-\frac{25}{7}$

42.  $4a^2 \sin^2 \left( \frac{3\pi}{4} \right) - 3[a \tan 225^\circ]^2 + [2a \cos 315^\circ]^2 =$

1) 0      2)  $a$       3)  $\sqrt{2}a$       4)  $a^2$

43. If  $\theta$  lies in the first quadrant and  $5 \tan \theta = 4$ , then

$\frac{5 \sin \theta - 3 \cos \theta}{\sin \theta + 2 \cos \theta} =$  (EAM-2007)

1)  $5/14$       2)  $3/14$       3)  $1/14$       4) 0

44. If  $a \sin^2 \theta + b \cos^2 \theta = c$ , then  $\tan^2 \theta =$

(EAM-2010)

1)  $\frac{b-c}{a-c}$       2)  $\frac{c-b}{a-c}$       3)  $\frac{a-c}{b-c}$       4)  $\frac{a-c}{c-b}$

### LEVEL-I(C.W)-KEY

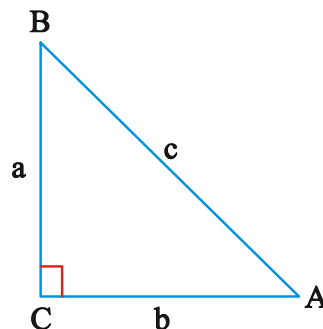
|       |       |       |       |       |
|-------|-------|-------|-------|-------|
| 01) 1 | 02) 2 | 03) 3 | 04) 2 | 05) 1 |
| 06) 2 | 07) 2 | 08) 4 | 09) 1 | 10) 4 |
| 11) 3 | 12) 2 | 13) 1 | 14) 2 | 15) 3 |
| 16) 1 | 17) 2 | 18) 2 | 19) 1 | 20) 3 |
| 21) 2 | 22) 1 | 23) 2 | 24) 3 | 25) 2 |
| 26) 1 | 27) 2 | 28) 2 | 29) 3 | 30) 1 |
| 31) 3 | 32) 2 | 33) 2 | 34) 3 | 35) 3 |
| 36) 2 | 37) 4 | 38) 4 | 39) 3 | 40) 1 |
| 41) 2 | 42) 4 | 43) 1 | 44) 2 |       |

LEVEL-I(C.W)-HINTS

1.  $-13 \times \frac{\pi}{6} \times \frac{180}{\pi} = -390^\circ$
2.  $1358^\circ = 8(180^\circ) - 82^\circ$ ,  
 $3608^\circ = 20(180^\circ) + 8^\circ$
3. Convert the corresponding values.
4.  $\sin 48^\circ \operatorname{cosec} 48^\circ + \cos 48^\circ \sec 48^\circ = 1 + 1 = 2$
5.  $\therefore \text{If } A + B = 90^\circ \Rightarrow \tan A \tan B = 1$
6.  $\cos(204^\circ) = \cos(180^\circ + 24^\circ)$   
 $\cos(125^\circ) = \cos(180^\circ - 55^\circ)$
7. Use  $(\sec^2 A - \tan^2 A) = 1$
8. Use  $(\sec^2 A - \tan^2 A) = 1$
9.  $A + B = 360^\circ \Rightarrow \sin A + \sin B = 0$
10. Put  $x = 90^\circ$
11. Put  $\alpha = 45^\circ$
12. Let  $x = 90^\circ$
13.  $A = B = C = 60^\circ$
14.  $1 = \operatorname{cosec}^2 \theta - \cot^2 \theta$
15.  $A + B = 90^\circ \Rightarrow \sin^2 A + \sin^2 B = 1$
16. If  $A + B = 90^\circ$  then  $(\cos^2 A + \cos^2 B) = 1$  and  
 $(\sin^2 A + \sin^2 B) = 1$
17.  $A + C = 180^\circ$ ,  $B + D = 180^\circ$
18. Simplify
19. Put  $\theta_1 = \theta_2 = \theta_3 = 90^\circ$
20. Divide the given equation by  $\sin^2 x$  both sides
21.  $\sin x = (1 - \sin^2 x) \Rightarrow \sin x = \cos^2 x$
22.  $x - y = \pi$
23. Put  $x = 90^\circ$
24.  $\alpha = \frac{\pi}{4}$
25.  $k^2 = (1 + \sin A)(1 + \sin B)(1 + \sin C)(1 - \sin A)$   
 $(1 - \sin B)(1 - \sin C)$   
 $K^2 = (1 - \sin^2 A)(1 - \sin^2 B)(1 - \sin^2 C)$   
 $K^2 = \cos^2 A \cdot \cos^2 B \cdot \cos^2 C$

$$K = \pm \cos A \cos B \cos C$$

26. Put  $A = 45^\circ$
27.  $\tan^2 \theta + \sec \theta = 5, \Rightarrow (\sec^2 \theta - 1) + \sec \theta = 5,$   
 $\Rightarrow \sec^2 \theta + \sec \theta - 6 = 0, \sec \theta = 2 \text{ or } -3$
28.  $\log(\sin 90^\circ) = 0$
29. Let  $(\alpha + \beta) = 60^\circ$   
 $(\alpha - \beta) = 45^\circ \Rightarrow 2\beta = 15^\circ \Rightarrow 6\beta = 45^\circ$
30. Square and add
31.  $\sin^2 A(\cos^2 B + \sin^2 B) + \cos^2 A(\sin^2 B + \cos^2 B)$  or  
Put  $A = B = 0^\circ$
32. Put  $\cot B = \frac{\cos B}{\sin B}$
33. use  $\sin^6 \theta + \cos^6 \theta = 1 - 3\sin^2 \theta \cos^2 \theta$
34. Substitute the corresponding values.
- 35.



$$\angle C = 90^\circ$$

$$\therefore \tan A + \tan B = \frac{a}{b} + \frac{b}{a} = \frac{a^2 + b^2}{ab} = \frac{c^2}{ab}$$

36.  $a = \sin 170^\circ + \cos 170^\circ = \sin 10^\circ - \cos 10^\circ < 0$
37.  $A + B = 90^\circ \Rightarrow \tan A \tan B = 1$
38.  $\frac{y}{x} = \frac{b \tan^3 \theta \sec \theta}{a \sec^3 \theta \tan \theta} = \frac{b}{a} \sin^2 \theta$
39. by squaring and adding  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 2$
40.  $\cot A \cot B \cot C > 0$   
 $\Rightarrow \cot A, \cot B, \cot C$  are positive  
 $\Rightarrow \triangle ABC$  is acute angled

41.  $\sin \alpha = \frac{3}{5}, \cos \beta = \frac{3}{5}, \cos \alpha = \frac{4}{5}, \sin \beta = \frac{4}{5}$
42. Given  $= 4a^2 \left( \frac{1}{\sqrt{2}} \right)^2 - 3[a \times 1]^2 + \left[ 2a \times \frac{1}{\sqrt{2}} \right]^2$
43. Divide the Nr. and Dr. with  $\cos \theta$
44. Divide throughout with  $\cos^2 \theta$   
 $\Rightarrow a \tan^2 \theta + b = c \sec^2 \theta \Rightarrow (a-c) \tan^2 \theta = (c-b)$



LEVEL - I (H.W)

1.  $-11\pi/3$  radians =  
 1)  $-390^\circ$  2)  $-620^\circ$  3)  $-610^\circ$  4)  $-660^\circ$
2.  $\sin 4530^\circ =$   
 1)  $\frac{1}{2}$  2)  $-\frac{1}{2}$  3)  $\frac{\sqrt{3}}{2}$  4)  $-\frac{\sqrt{3}}{2}$
3.  $\frac{\sin 150^\circ - 5 \cos 300^\circ + 7 \tan 225^\circ}{\tan 135^\circ + 3 \sin 210^\circ} =$   
 1) 10 2) -5 3) -2 4)  $-3/2$
4.  $\cos 23^\circ \operatorname{cosec} 67^\circ - \sin 23^\circ \sec 67^\circ =$   
 1) 0 2) 2 3) 1 4) -1
5.  $\log \tan 18^\circ + \log \tan 36^\circ + \log \tan 54^\circ + \log \tan 72^\circ =$   
 1)  $\log 4$  2)  $\log 3$  3)  $\log 2$  4) 0
6.  $\cos 5^\circ + \cos 24^\circ + \cos 175^\circ + \cos 204^\circ + \cos 300^\circ =$   
 1)  $1/2$  2) 1 3)  $3/2$  4) 2
7.  $\operatorname{cosec} A + \cot A = \frac{2}{3} \Rightarrow \cos A =$   
 1)  $\frac{5}{13}$  2)  $\frac{13}{5}$  3)  $-\frac{5}{13}$  4)  $-\frac{13}{5}$
8. If  $\operatorname{cosec} \theta - \cot \theta = 5$  then  $\theta$  lies in the quadrant  
 1) I 2) II 3) III 4) IV
9.  $\tan 20^\circ + \tan 40^\circ + \tan 60^\circ + \dots + \tan 180^\circ =$   
 1) 0 2) 1 3) 2 4) 3
10.  $(\sin \alpha + \operatorname{cosec} \alpha)^2 + (\sec \alpha + \cos \alpha)^2 =$   
 $k + \tan^2 \alpha + \cot^2 \alpha \Rightarrow k =$   
 1) 9 2) 7 3) 5 4) 3
11.  $\frac{1}{(1 + \cot^2 \alpha)^2} + \frac{\tan^2 \alpha}{(1 + \tan^2 \alpha)^2} + \frac{1}{1 + \tan^2 \alpha} =$   
 1) -1 2) 0 3) 1 4) 2

12.  $\frac{\cos^3 A + \sin^3 A}{\cos A + \sin A} + \frac{\cos^3 A - \sin^3 A}{\cos A - \sin A} = k$  then  $k =$   
 1) 0 2) 1 3) 2 4) -1
13. In  $\triangle ABC$ ,  $\tan(A - B - C) =$   
 1)  $\sin 2A$  2) 1 3)  $\tan 2A$  4) 0
14.  $\frac{1 + \cot \alpha + \operatorname{cosec} \alpha}{1 - \cot \alpha + \operatorname{cosec} \alpha} =$   
 1)  $\frac{\sin \alpha}{1 + \cos \alpha}$  2)  $\frac{\sin \alpha}{1 - \cos \alpha}$   
 3)  $\frac{1 + \cos \alpha}{\sin \alpha}$  4)  $\frac{1 - \sin \alpha}{\cos \alpha}$
15.  $\cos^2(125^\circ - x) - \cos^2(55^\circ + x) =$   
 1) -1 2) 0 3) 1 4) 2
16.  $\sin^2 \frac{\pi}{18} + \sin^2 \frac{2\pi}{18} + \sin^2 \frac{4\pi}{18} + \sin^2 \frac{8\pi}{18}$   
 $+ \sin^2 \frac{7\pi}{18} + \sin^2 \frac{5\pi}{18} =$   
 1) 1 2) 2 3) 3 4) 4
17. If A B C D is a quadrilateral then  
 $\tan\left(\frac{A+B}{4}\right) =$   
 1)  $\cos\left(\frac{C-D}{4}\right)$  2)  $\cot\left(\frac{C-D}{4}\right)$   
 3)  $\cos\left(\frac{C+D}{4}\right)$  4)  $\cot\left(\frac{C+D}{4}\right)$
18.  $\left[ \cos\left(\frac{\pi}{2} - x\right) + \cos(\pi - x) \right]^2 + \left[ \sin\left(\frac{3\pi}{2} - x\right) + \sin(2\pi - x) \right]^2 =$   
 1) 0 2) 2 3) 1 4) 4
19. If  $\cos \theta_1 + \cos \theta_2 + \cos \theta_3 + \cos \theta_4 + \cos \theta_5 = 5$   
 then  $\sin \theta_1 + \sin \theta_2 + \sin \theta_3 + \sin \theta_4 + \sin \theta_5 =$   
 1) 3 2) 2 3) 1 4) 0
20.  $9 \cos^2 x + 4 \sin^2 x = 5 \Rightarrow \tan x =$   
 1)  $\pm 1$  2)  $\pm 2$  3)  $\pm 3$  4)  $\pm 4$
21.  $\cos x + \cos^2 x = 1 \Rightarrow \sin^8 x + 2 \sin^6 x + \sin^4 x =$   
 1) 0 2) 1 3) 2 4) -1
22. If  $\cot(\alpha + \beta) = 0$  then  $\sin(\alpha + 2\beta) =$   
 1)  $\sin \alpha$  2)  $\cos \alpha$  3)  $\sin \beta$  4)  $\cos \beta$

23. If  $\tan \alpha + \cot \alpha = 2$  then  $\sqrt{\tan \alpha} + \sqrt{\cot \alpha} =$   
 1)  $\sqrt{2}$  2)  $2\sqrt{2}$  3) 2 4)  $4\sqrt{2}$
24.  $5 \sin x + 4 \cos x = 3 \Rightarrow 4 \sin x - 5 \cos x =$   
 1) 4 2)  $4\sqrt{2}$  3)  $3\sqrt{2}$  4)  $\sqrt{2}$
25. If  $k = (\sec A + \tan A)(\sec B + \tan B)(\sec C + \tan C) =$   
 $(\sec A - \tan A)(\sec B - \tan B)(\sec C - \tan C)$   
 then  $k =$   
 1) 0 2)  $\pm 1$  3)  $\pm 3$  4)  $\pm 4$
26. The value of  $\cot^2 \alpha \left( \frac{\sec \alpha - 1}{1 + \sin \alpha} \right)$   
 $+ \sec^2 \alpha \left( \frac{\sin \alpha - 1}{1 + \sec \alpha} \right)$  is  
 1) 0 2) 1 3) 2 4) -2
27.  $\sec^2 \theta + \tan \theta = 13 \Rightarrow \tan \theta =$   
 1) 4 2) 3 3) -3 4) 5
28.  $\log(\tan 1^\circ) \log(\tan 2^\circ) \log(\tan 3^\circ) \dots \log(\tan 45^\circ) =$   
 1) 0 2) 1 3) -1 4)  $\frac{1}{2}$
29. If  $\sin(\alpha + \beta) = 1$ ,  $\sin(\alpha - \beta) = \frac{1}{2}$  then  
 $\tan(\alpha + 2\beta) \tan(2\alpha + \beta) =$   
 1) 1 2) -1 3) 0 4) 2
30. If  $x = r \cos \alpha \cos \beta \cos \gamma$ ;  
 $y = r \cos \alpha \cos \beta \sin \gamma$ ;  
 $z = r \sin \alpha \cos \beta$ ;  $\mu = r \sin \beta$  then  
 $x^2 + y^2 + z^2 + \mu^2 =$   
 1)  $r$  2)  $2r$  3)  $r^2$  4)  $4r^2$
31.  $\sec^2 A \sec^2 B - \sec^2 A \tan^2 B - \tan^2 A \sec^2 B + \tan^2 A \tan^2 B =$   
 1) 1 2) 0 3) -1 4) 2
32.  $2 \cos^2 B - 1 = \tan^2 A$ , then  $\cos A \cos B =$   
 1)  $\pm \frac{1}{2}$  2)  $\pm \frac{1}{3}$  3)  $\pm \frac{1}{\sqrt{2}}$  4)  $\pm 1$
33.  $\sin^4 \theta + 2 \sin^2 \theta \left( 1 - \frac{1}{\cos^2 \theta} \right) + \cos^4 \theta =$   
 1) 1 2) 0 3)  $\frac{1}{2}$  4) -1
34. If  $x \cot^2 120^\circ + 4 \sin^2 150^\circ = 3$ , then  $x =$   
 1) 2 2) 7 3) 6 4) 5

35. In a right angled triangle ABC,  $\angle C = 90^\circ$ , then  
 $\cos^2 A + \cos^2 B =$   
 1) 2 2) 1 3)  $\frac{1}{2}$  4)  $\frac{3}{4}$
36. If  $x = \sin 130^\circ + \cos 130^\circ$  then  
 1)  $x < 0$  2)  $x = 0$  3)  $x > 0$  4)  $x \geq 0$
37.  $\cot \frac{\pi}{20} \cdot \cot \frac{3\pi}{20} \cdot \cot \frac{5\pi}{20} \cdot \cot \frac{7\pi}{20} \cdot \cot \frac{9\pi}{20} \cdot \cot \frac{15\pi}{20} =$   
 1) 1 2) -1 3)  $\sqrt{3}$  4)  $-\sqrt{3}$
38. If  $x = a \sec^n \theta$ ;  $y = b \tan^n \theta$  then  
 $\left( \frac{x}{a} \right)^{2/n} - \left( \frac{y}{b} \right)^{2/n} =$   
 1) 0 2) -1 3) 1 4) 2
39. If  $x = h + p \sec \alpha$ ,  $y = k + q \csc \alpha$  then  
 $\left( \frac{p}{x-h} \right)^2 + \left( \frac{q}{y-k} \right)^2 =$   
 1) 1 2) -1 3) 0 4)  $1/2$
40. If A, B, C are angles of a triangle such that A is obtuse then  
 1)  $\tan A \tan B > 1$  2)  $\tan B \tan C < 1$   
 3)  $\tan C \tan A > 1$  4)  $\tan A \tan B \tan C > 1$
41. If  $\alpha, \beta$  are complementary angles,  
 $\sin \alpha = \frac{3}{5}$ , then  $\cos \alpha \cos \beta - \sin \alpha \sin \beta =$   
 1) 1 2) 2 3)  $\frac{4}{5}$  4) 0
42.  $a^2 \cos^2 \frac{2\pi}{3} - 4a^2 \tan^2 \frac{3\pi}{4} + 2a^2 \sin^2 \frac{2\pi}{3} =$   
 1)  $a^2$  2) 0 3)  $\frac{3a^2}{4}$  4)  $\frac{-9a^2}{4}$
43. If  $\tan \theta = \frac{p}{q}$  then  $\frac{p \sin \theta - q \cos \theta}{p \sin \theta + q \cos \theta} =$   
 1)  $\frac{2p}{p^2 + q^2}$  2)  $\frac{2pq}{p^2 + q^2}$   
 3)  $\frac{p^2 - q^2}{p^2 + q^2}$  4)  $\frac{q^2 - p^2}{p^2 + q^2}$



44 If  $a \sec \theta + b \tan \theta = c$  then

$$(a \tan \theta + b \sec \theta)^2 =$$

- 1)  $a^2 + b^2 + c^2$       2)  $-a^2 + b^2 + c^2$   
3)  $a^2 - b^2 + c^2$       4)  $a^2 + b^2 - c^2$ .

**LEVEL-I-(H.W)-KEY**

- 01) 4    02) 2    03) 3    04) 1    05) 4    06) 1  
07) 3    08) 2    09) 1    10) 2    11) 3    12) 3  
13) 3    14) 3    15) 2    16) 3    17) 4    18) 2  
19) 4    20) 2    21) 2    22) 4    23) 3    24) 2  
25) 2    26) 1    27) 2    28) 1    29) 1    30) 3  
31) 1    32) 3    33) 1    34) 3    35) 2    36) 3  
37) 2    38) 3    39) 1    40) 2    41) 4    42) 4  
43) 3    44) 2

**LEVEL-I-(H.W)-HINTS**

- $-11 \frac{\pi}{3} \times \frac{180}{\pi} = -660^\circ$
- $\sin(4530^\circ) = \sin((12)360^\circ + 210^\circ)$   
 $= \sin 210^\circ = -\sin 30^\circ = -\frac{1}{2}$
- Convert the corresponding values.
- $\cos 23^\circ \operatorname{cosec}(90^\circ - 23^\circ) - \sin 23^\circ \sec(90^\circ - 23^\circ)$   
 $= 1 - 1 = 0$
- $\log(\tan 18^\circ \tan 36^\circ \tan 54^\circ \tan 72^\circ) = \log 1 = 0$   
[ $\because$  If  $A + B = 90^\circ \Rightarrow \tan A \tan B = 1$ ]
- $\alpha + \beta = 180^\circ$   
 $\Rightarrow \cos \alpha + \cos \beta = 0$
- Use  $(\cos^2 \theta - \cot^2 \theta) = 1$
- Use  $(\cos^2 A - \cot^2 A) = 1$ . Find the quadrant in which  $A$  lies
- $A + B = 180^\circ \Rightarrow \tan A + \tan B = 0$
- Square and simplify
- Put  $\alpha = 45^\circ$
- Put  $A = 0^\circ$
- $B + C = 180^\circ - A \Rightarrow \tan(A + A - 180^\circ) = \tan 2A$
- Use  $1 = (\operatorname{cosec}^2 \alpha - \cot^2 \alpha)$
- $(125^\circ - x) + (55^\circ + x) = 180^\circ$
- If  $(A + B) = \frac{\pi}{2}$  then  $\sin^2 A + \sin^2 B = 1$

$$17. (A + B + C + D) = 360^\circ \Rightarrow \frac{A+B}{4} = \frac{\pi}{2} - \frac{C+D}{4}$$

$$18. (\sin x - \cos x)^2 + (-\cos x - \sin x)^2 = 2$$

$$19. \text{Put } \theta_1 = \theta_2 = \theta_3 = 0^\circ$$

20. Divide the given equation by  $\cos^2 x$  both sides.

$$21. \cos x = 1 - \cos^2 x \Rightarrow \cos x = \sin^2 x$$

$$22. \cot(\alpha + \beta) = 0 \Rightarrow \alpha + \beta = \frac{\pi}{2}$$
  
$$\Rightarrow \sin(\alpha + 2\beta) = \sin(90^\circ + \beta) = \cos \beta$$

$$23. \text{Put } \alpha = \frac{\pi}{4}$$

$$24. a \sin x + b \cos x = c \Rightarrow b \sin x - a \cos x = \pm \sqrt{a^2 + b^2 - c^2}$$

$$25. K^2 = (1 + \sin A)(1 + \sin B)(1 + \sin C)(1 - \sin A)(1 - \sin B)(1 - \sin C)$$
  
$$K^2 = (1 - \sin^2 A)(1 - \sin^2 B)(1 - \sin^2 C)$$
  
$$K^2 = \cos^2 A \cdot \cos^2 B \cdot \cos^2 C$$
  
$$K = \pm \cos A \cos B \cos C$$

26. Put  $\alpha = 45^\circ$ ; verify the options.

$$27. 1 + \tan^2 \theta + \tan \theta = 13 \Rightarrow \tan \theta = 3 \text{ or } -4$$

$$28. \log(\tan 45^\circ) = \log 1 = 0$$

$$29. \alpha + \beta = 90^\circ, \alpha - \beta = 30^\circ \Rightarrow \alpha = 60^\circ, \beta = 30^\circ$$

30. Squaring and adding

$$31. \text{Put } A = B = 45^\circ$$

$$32. 2 \cos^2 B = 1 + \tan^2 A$$

$$33. \sin^4 \theta + \cos^4 \theta = 1 - 2 \sin^2 \theta \cos^2 \theta$$

$$34. x \cdot \frac{1}{3} + 4 \cdot \frac{1}{4} = 3 \Rightarrow x = 6$$

$$35. A + B = 90^\circ \Rightarrow \cos^2 A + \cos^2 B = 1$$

$$36. x = \sin 50^\circ - \cos 50^\circ$$

$$37. A + B = 90^\circ \Rightarrow \cot A \cot B = 1$$

$$38. \sec^2 \theta - \tan^2 \theta = 1$$

$$39. \sin^2 \theta + \cos^2 \theta = 1$$

$$40. A \text{ is obtuse } \Rightarrow B + C < 90^\circ \Rightarrow \tan B \tan C < 1$$

$$41. \sin \alpha = \frac{3}{5}, \cos \beta = \frac{3}{5}, \cos \alpha = \frac{4}{5}, \sin \beta = \frac{4}{5}$$

$$42. a^2 \left(-\frac{1}{2}\right)^2 - 4a^2 (-1)^2 + 2a^2 \left(\frac{\sqrt{3}}{2}\right)^2$$

43. Divide the Nr and Dr. with  $\cos \theta$

44. Let  $a \tan \theta + b \sec \theta = k$

$$\therefore a^2 - b^2 = c^2 - k^2 \Rightarrow k^2 = c^2 - a^2 + b^2$$



1. If  $\theta$  is not in 4th quadrant and  $\tan \theta = -4/3$  then

$$5 \sin \theta + 10 \cos \theta + 9 \sec \theta + 16 \operatorname{cosec} \theta + 4 \cot \theta =$$

1) -1      2) 2/5      3) 4/5      4) 0

2.  $f(x) = x^3 - 3x + 5$  then  $f\left(\sin \frac{3\pi}{2}\right) + f\left(\cos \frac{3\pi}{2}\right) =$

1) 10      2) 12      3) 14      4) 16

3. If  $\tan \theta, 2 \tan \theta + 2, 3 \tan \theta + 3$  are in G.P then

$$\text{the value of } \frac{7 - 5 \cot \theta}{9 - 4\sqrt{\sec^2 \theta - 1}} \text{ is}$$

1)  $\frac{12}{5}$       2)  $\frac{-33}{28}$       3)  $\frac{33}{100}$       4)  $\frac{12}{13}$

4. In a triangle ABC,  $C = 90^\circ$ , then the equation whose roots are  $\tan A, \tan B$  is

1)  $abx^2 + c^2x + ab = 0$

2)  $abx^2 + c^2x - ab = 0$

3)  $abx^2 - c^2x - ab = 0$

4)  $abx^2 - c^2x + ab = 0$

5. Which of the following is correct?

1)  $\sin 1^\circ > \sin 1$       2)  $\sin 1^\circ < \sin 1$

3)  $\sin 1^\circ = \sin 1$       4)  $\sin 1^\circ = \frac{\pi}{180} \sin 1$

6. If  $a = \cos 3$  and  $b = \sin 8$  then

1)  $a > 0, b > 0$       2)  $ab < 0$       3)  $a > b$       4)  $ab > 0$

7.  $\operatorname{cosec} A = 4p + \frac{1}{16p} \Rightarrow \operatorname{cosec} A + \cot A =$

1)  $8p$       2)  $\frac{1}{8p}$       3)  $-8p$  (or)  $\frac{1}{8p}$       4)  $8p$  (or)  $\frac{1}{8p}$

8. If  $\sqrt{\frac{1 - \sin A}{1 + \sin A}} = \sec A - \tan A$  then A lies in the quadrants

1) I, II      2) II, III      3) I, IV      4) I, III

9. If  $\sin \theta + \cos \theta = m$  and  $\sec \theta + \operatorname{cosec} \theta = n$ , then  $n(m+1)(m-1) =$

1) m      2) n      3)  $2m$       4)  $2n$

10. If  $1 + \sin x + \sin^2 x + \dots$  to  $\infty = 4 + 2\sqrt{3}, 0 < x < \pi$

and  $x \neq \frac{\pi}{2}$  then  $x =$

1)  $30^\circ, 60^\circ$       2)  $60^\circ, 120^\circ$

3)  $90^\circ, 120^\circ$       4)  $30^\circ, 45^\circ$

11.  $\cos^2 5^\circ + \cos^2 10^\circ$

$+ \cos^2 15^\circ + \dots + \cos^2 360^\circ =$

1) 18      2) 27      3) 36      4) 45

12. If  $x = a \cos^2 \theta \sin \theta$  and  $y = a \sin^2 \theta \cos \theta$ ,

then  $\frac{(x^2 + y^2)^3}{x^2 y^2} =$

1) a      2)  $a^3$       3)  $a^2$       4)  $a^5$

13.  $a \sec \theta + b \tan \theta = 1, a \sec \theta - b \tan \theta = 5$

$\Rightarrow a^2(b^2 + 4) =$

1)  $3b^2$       2)  $9b^2$       3)  $b^2$       4)  $4b^2$

14.  $\tan A = a \tan B, \sin A = b \sin B \Rightarrow \frac{b^2 - 1}{a^2 - 1} =$

1)  $\sin^2 A$       2)  $\sin^3 A$       3)  $\cos^2 A$       4)  $\cos^3 A$

15. If  $\frac{\cos^2 \theta}{a} = \frac{\sin^2 \theta}{b}$  then  $\frac{\cos^4 \theta}{a} + \frac{\sin^4 \theta}{b} =$

1)  $\frac{1}{a+b}$       2)  $\frac{1}{(a+b)^2}$

3)  $\frac{1}{a^2} + \frac{1}{b^2}$       4)  $a+b$

16. If  $a = x \cos^2 A + y \sin^2 A$  then  $(x-a)(y-a)$

$+ (x-y)^2 \sin^2 A \cos^2 A =$

1) 0      2) 1      3)  $xy + a^2$       4)  $xy - a^2$

17. If  $a \cos^3 \alpha + 3a \cos \alpha \sin^2 \alpha = m$  and  $a \sin^3 \alpha + 3a \cos^2 \alpha \sin \alpha = n$  then

$(m+n)^{2/3} + (m-n)^{2/3} =$

1)  $2a^2$       2)  $2a^{1/3}$       3)  $2a^{2/3}$       4)  $2a^3$

18.  $\cos^4 \alpha - \sin^4 \alpha = a$  then  $\frac{1-a}{1+a} =$

1)  $\tan^2 \alpha$       2)  $\cot^2 \alpha$       3)  $-\tan^2 \alpha$       4)  $-\cot^2 \alpha$

19.  $a = \frac{1 + \sin x}{1 - \cos x + \sin x} \Rightarrow \frac{1 + \cos x + \sin x}{2 \sin x} =$

- 1)  $a$       2)  $\frac{1}{a}$       3)  $a^2$       4)  $\frac{1}{a^2}$

20. Eliminate  $\theta$  from  $x = 1 + \tan \theta$ ,  $y = 2 + \cot \theta$

- 1)  $xy + 1 = x + y$       2)  $xy + 2 = 2x + y$   
3)  $xy + 1 = 2x + y$       4)  $xy + 1 = 2y + x$

21.  $f(x) = \sin^2 x + \cos^2 x \Rightarrow$

- 1)  $f(x) < 1$       2)  $f(x) = 1$   
3)  $1 < f(x) < 2$       4)  $f(x) \geq 2$

22. Which of the following is not possible?

- 1)  $\sin \theta = \frac{5}{7}$       2)  $\cos \theta = \frac{1 + a^2}{1 - a^2}, |a| \neq 1$   
3)  $\tan \theta = 100$       4)  $\sec \theta = \frac{5}{2}$

23. If  $\theta = \frac{11\pi}{6}$ , then  $\cos \theta + \sin \theta =$

- 1)  $\frac{\sqrt{3} + 1}{\sqrt{2}}$       2)  $\frac{\sqrt{3} - 1}{\sqrt{2}}$       3)  $\frac{\sqrt{3} - 1}{2}$       4)  $\frac{\sqrt{3} + 1}{2}$

24. The value of  $\sin\left(n\pi + (-1)^n \frac{\pi}{4}\right)$ ,  $n \in I$ , is

- 1) 0      2)  $\frac{1}{\sqrt{2}}$       3)  $\frac{-1}{\sqrt{2}}$       4)  $\frac{\sqrt{3}}{2}$

25. If  $e^{(\sin^2 x + \sin^4 x + \sin^6 x + \dots \infty) \log 2} = 8$  and

$0 < x < \frac{\pi}{2}$  then  $\frac{\cos x}{\cos x + \sin x} =$

- 1)  $\frac{\sqrt{3} + 1}{2}$       2)  $\frac{\sqrt{3} - 1}{2}$       3)  $\frac{2}{\sqrt{3} + 1}$       4)  $\frac{2}{\sqrt{3} - 1}$

26. If  $\cos x = \tan y$ ,  $\cot y = \tan z$  and  $\cot z = \tan x$ , then  $\sin x =$  (EAM-20014)

- 1)  $\frac{\sqrt{5} + 1}{4}$       2)  $\frac{\sqrt{5} - 1}{4}$       3)  $\frac{\sqrt{5} + 1}{2}$       4)  $\frac{\sqrt{5} - 1}{2}$

27. If  $\sin \theta + \cos \theta = p$  and  $\sin^3 \theta + \cos^3 \theta = q$ , then  $p(p^2 - 3) =$  (EAM-2013)

- 1)  $q$       2)  $2q$       3)  $-q$       4)  $-2q$

28. If  $0 \leq x \leq \pi$ ,  $81^{\sin^2 x} + 81^{\cos^2 x} = 30$  then  $x =$

- 1)  $\frac{\pi}{6}$       2)  $\frac{\pi}{4}$       3)  $\frac{\pi}{15}$       4)  $\frac{\pi}{8}$

29.  $\cos \sec^2 \alpha = \frac{4xy}{(x+y)^2}$   $[(x, y) \neq (0, 0)]$  is

- 1) Possible for  $x = y$       2) Impossible for  $x = y$   
3) Possible for  $x = -y$       4) Not possible

30. If  $a \sin \theta + b \cos \theta = c$  then  $\frac{a - b \tan \theta}{b + a \tan \theta} =$

- 1)  $\frac{\pm \sqrt{b^2 - c^2 + a^2}}{c}$       2)  $\frac{\pm \sqrt{b^2 + c^2 - a^2}}{c}$   
3)  $\frac{\pm \sqrt{b^2 - c^2 - a^2}}{c}$       4)  $\frac{\pm \sqrt{a^2 + b^2 + c^2}}{b^2 - c^2}$

### LEVEL-II(C.W)-KEY

- 01) 4    02) 2    03) 2    04) 4    05) 2    06) 2  
07) 4    08) 3    09) 3    10) 2    11) 3    12) 3  
13) 2    14) 3    15) 1    16) 1    17) 3    18) 1  
19) 1    20) 3    21) 4    22) 2    23) 3    24) 2  
25) 2    26) 4    27) 4    28) 1    29) 1    30) 1

### LEVEL-II(C.W)-HINTS

- $\theta$  lies in II quadrant, substitute the values
- $\sin 270^\circ = -1$  &  $\cos 270^\circ = 0$
- $(2 \tan \theta + 2)^2 = \tan \theta (3 \tan \theta + 3)$   
 $\Rightarrow \tan^2 \theta + 5 \tan \theta + 4 = 0 \Rightarrow \tan \theta = -1, -4$   
but  $\tan \theta = -4$  only satisfies above condition
- $\tan A = \frac{a}{b}$ ,  $\tan B = \frac{b}{a}$  and  $(a^2 + b^2) = c^2$  Find sum and product of roots
- $1 > 1^0$
- $3^\circ$  lies in the second quadrant  $\therefore \cos 3 < 0$   
 $8^\circ$  lies in the second quadrant  $\therefore \sin 8 > 0$   
 $\Rightarrow ab < 0$
- Use  $(\cos \sec^2 A - \cot^2 A) = 1$   
If  $\cos \sec A = \left(4P + \frac{1}{16P}\right)$   
then  $\cot A = \pm \left(4P - \frac{1}{16P}\right)$

8.  $\sqrt{\cos^2 A} = \cos A$  then  $A \in Q_1, Q_4$
9. Find  $m^2$  and multiply
10. Use formula  $S_\infty = \frac{a}{1-r}$  for G.P
11. If  $(A+B) = 360^\circ$  or  $(A+B) = 180^\circ$   
then  $\cos^2 A = \cos^2 B$  and  $(A+B) = 90^\circ$  then  
 $(\cos^2 A + \cos^2 B) = 1$
12.  $\theta$  can be eliminated by substitution.
13. Given  $(a \sec \theta + b \tan \theta) = 1$   
 $\Rightarrow (a \sec \theta - b \tan \theta) = 5$ , eliminate  $\theta$
14. Put  $A = 30^\circ$  and  $B = 45^\circ$  and verify
15. Find  $\tan \theta$  and simplify
16. Put  $A = 90^\circ$
17.  $m+n = a(\cos \alpha + \sin \alpha)^3$   
 $m-n = a(\cos \alpha - \sin \alpha)^3$
18.  $a = \cos^2 \alpha - \sin^2 \alpha$   
 $\Rightarrow 1-a = 2 \sin^2 \alpha$  and  $1+a = 2 \cos^2 \alpha$
19. In R.H.S. of given equation multiply and divide by  
 $(1 + \cos x + \sin x)$
20.  $x-1 = \tan \theta, y-2 = \cot \theta$   
Multiply two equations
21.  $f(x) = \left( \sin^2 x + \frac{1}{\sin^2 x} \right) \geq 2$   
 $\therefore \left( a + \frac{1}{a} \right) \geq 2$  for any real number  $a > 0$
22. for  $a = 2, \cos \theta = \frac{5}{-3}$  not possible
23. Put  $\theta = 330^\circ$  substitution
24. Put  $n = 0$
25.  $\sin^2 x + \sin^4 x + \dots \infty = \frac{\sin^2 x}{1 - \sin^2 x} = \tan^2 x$   
 $\therefore (e^{\log_e 2})^{\tan^2 x} = 8 \Rightarrow 2^{\tan^2 x} = 2^3 \Rightarrow \tan x = \sqrt{3}$   
 $\therefore \text{G.E} = \frac{1}{1 + \tan x} = \frac{1}{1 + \sqrt{3}} = \frac{\sqrt{3}-1}{2}$

26.  $\left. \begin{array}{l} \cos x = \tan y \\ \tan z = \cot y \\ \cot z = \tan x \end{array} \right\} \text{multiplying we get } \cos x = \tan x$   
 $\Rightarrow \cos^2 x = \sin x$   
 $\Rightarrow \sin^2 x + \sin x - 1 = 0 \Rightarrow \sin x = \frac{-1 \pm \sqrt{5}}{2}$
27.  $\cos \theta + \sin \theta = p \Rightarrow p^2 = 1 + 2 \sin \theta \cos \theta \Rightarrow \sin \theta \cos \theta = \frac{p^2 - 1}{2}$   
also  
 $p^3 = \cos^3 \theta + \sin^3 \theta + 3 \cos \theta \sin \theta (\cos \theta + \sin \theta)$   
 $\Rightarrow p^3 = q + 3 \left( \frac{p^2 - 1}{2} \right) p$   
 $\Rightarrow 2p^3 = 2q + 3p^3 - 3p \Rightarrow p^3 - 3p = -2q$
28. If  $81^{\sin^2 x} = y \Rightarrow 81^{\cos^2 x} = \frac{81}{y}$   
 $\therefore y + \frac{81}{y} = 30 \Rightarrow y^2 - 30y + 81 = 0 \Rightarrow y = 27, 3$
29.  $\therefore \operatorname{cosec} \alpha \in (-\infty, -1] \cup [1, \infty)$
30.  $G.E = \frac{a \cos \theta - b \sin \theta}{a \sin \theta + b \cos \theta} = \frac{\pm \sqrt{b^2 - c^2 + a^2}}{c}$

 **LEVEL - II (H.W)**

1. If  $\cos \theta = \frac{3}{5}$  and  $\theta$  is not in the first quadrant,  
then  $\frac{5 \tan(\pi + \theta) + 4 \cos(\pi + \theta)}{5 \sec(2\pi - \theta) - 4 \cot(2\pi + \theta)} =$   
1)  $\frac{4}{5}$       2)  $-\frac{4}{5}$       3)  $\frac{5}{4}$       4)  $-\frac{5}{4}$
2.  $f(x) = x^3 - 2x^2 + 3x - 5$   
 $\Rightarrow f\left[\sin\left(\frac{5\pi}{2}\right)\right] + f\left[\sin\left(\frac{3\pi}{2}\right)\right] =$   
1) 10      2) -10      11) 14      4) -14
3.  $\cos A, \sin A, \cot A$  are in GP  
then  $\tan^6 A - \tan^2 A =$   
1) -1      2) 0      3) 1      4) 2

4. If  $ABCD$  is a cyclic quadrilateral such that  $12 \tan A - 5 = 0$  and  $5 \cos B + 3 = 0$ , then  $\cos C \tan D =$

1)  $\frac{-16}{13}$  2)  $\frac{16}{13}$  3)  $\frac{-13}{16}$  4)  $\frac{23}{16}$

5.  $x = \cos 1^\circ, y = \cos 1 \Rightarrow$

1)  $x = y$  2)  $x > y$  3)  $x < y$  4)  $2x = y$

6.  $a = \sec 2^\circ, b = \sec 2 \Rightarrow$

1)  $a = b$  2)  $a < b$  3)  $b < a$  4)  $2a = b$

7.  $\tan \theta = P - \frac{1}{4p} \Rightarrow \sec \theta - \tan \theta =$

1)  $2p$  (or)  $\frac{1}{2p}$  2)  $\frac{1}{2p}$  (or)  $-2p$

3)  $-\frac{1}{2p}$  (or)  $2p$  4)  $-\frac{1}{2p}$  (or)  $-2p$

8. If  $\sqrt{\frac{1+\sin A}{1-\sin A}} = \sec A + \tan A$ , then  $A$  lies in the quadrants

1) I, II 2) II, III 3) I, III 4) I, IV

9.  $x = \tan \theta + \cot \theta, y = \cos \theta - \sin \theta \Rightarrow$

1)  $x = y$  2)  $\frac{1-y^2}{2} = \frac{1}{x}$

3)  $\frac{y^2-1}{2} = \frac{1}{x}$  4)  $\frac{1+y^2}{2} = \frac{1}{x}$

10.  $1 + \cos x + \cos^2 x + \dots$  to  $\infty = 4 + 2\sqrt{3}$  then  $x =$

1)  $30^\circ$  2)  $60^\circ$  3)  $45^\circ$  4)  $90^\circ$

11.  $\sin^2 5^\circ + \sin^2 10^\circ + \dots + \sin^2 180^\circ =$

1) 18 2) 27 3) 1 4) 0

12. If  $x = a \cos^3 \theta \sin^2 \theta, y = a \sin^3 \theta \cos^2 \theta$  and

$\frac{(x^2 + y^2)^p}{(xy)^q}, (p, q \in N)$  is independent of  $\theta$

then

1)  $p + q = 6$  2)  $4p = 5q$

3)  $4q = 5p$  4)  $pq = 16$

13.  $a = \sec \theta - \tan \theta, b = \operatorname{cosec} \theta + \cot \theta \Rightarrow a =$

1)  $\frac{b+1}{b-1}$  2)  $\frac{1+b}{1-b}$  3)  $\frac{b-1}{b+1}$  4)  $\frac{1-b}{1+b}$

14.  $\cos A = a \cos B, \sin A = b \sin B \Rightarrow (b^2 - a^2) \sin^2 B =$

1)  $1 + a^2$  2)  $2 + a^2$  3)  $1 - a^2$  4)  $2 - a^2$

15. If  $\frac{\sin x}{a} = \frac{\cos x}{b} = \frac{\tan x}{c} = k$  then

$bc + \frac{1}{ck} + \frac{ak}{1+bk}$  is

1)  $k \left( a + \frac{1}{a} \right)$  2)  $\frac{1}{k} \left( a + \frac{1}{a} \right)$  3)  $\frac{1}{k^2}$  4)  $\frac{a}{k}$

16. If  $m \cos^2 A + n \sin^2 A = p$ , then  $\cot^2 A =$

1)  $\frac{p+n}{m+p}$  2)  $\frac{p-n}{p-m}$  3)  $\frac{p-n}{m-p}$  4)  $\frac{n+1}{m+p}$

17. If  $a \sin^3 x + b \cos^3 x = \sin x \cos x$

and  $a \sin x = b \cos x$  then  $a^2 + b^2 =$

1) 0 2) 1 3) 2 4) 3

18. If  $\sin \theta + \cos \theta = a$ , then  $\sin^4 \theta + \cos^4 \theta =$

1)  $1 - \frac{1}{2}(a^2 + 1)^2$  2)  $1 - \frac{1}{2}(a^2 - 1)^2$

3)  $1 + \frac{1}{2}(a^2 + 1)^2$  4)  $1 + \frac{1}{2}(a^2 - 1)^2$

19. If  $\frac{\sin \theta - \cos \theta + 1}{\sin \theta + \cos \theta - 1} = \frac{x}{\tan \theta - \sec \theta + 1}$  then  $x =$

1) 0 2) 2  
3)  $\tan \theta - \sec \theta + 1$  4)  $\tan \theta + \sec \theta - 1$

20. If  $\tan \theta + \sin \theta = m$ ,  $\tan \theta - \sin \theta = n$ , then

$(m^2 - n^2)^2 =$

1)  $16mn$  2)  $4mn$  3)  $32mn$  4)  $8mn$

21. If  $\sec \theta + \cos \theta = 2$  then  $\sin^2 \theta + \tan^2 \theta =$

1) 0 2) 1 3) 2 4) 4

22. Which of the following is possible?

1)  $\sin \theta = 2$  2)  $\cos \theta = \frac{17}{5}$

3)  $\tan \theta = 2$  4)  $\sec \theta = \frac{4}{5}$

23. If  $\theta = \frac{\pi}{21}$ , then  $\frac{\sin 23\theta - \sin 7\theta}{\sin 2\theta + \sin 14\theta} =$

1) 0 2) 1 3) -1 4) 2

24. The value of  $\tan\left(n\pi + \frac{\pi}{3}\right), n \in I$  is  
 1)  $\sqrt{3}$       2)  $\frac{1}{\sqrt{3}}$       3) 0      4)  $\frac{1}{\sqrt{2}}$
25. If  $e^{(1+\sin^2 x + \sin^4 x + \dots \infty) \log 2} = 16$ , then  
 $\tan^2 x =$   
 1) 1      2) 2      3) 3      4) 4
26. If  $2 \sin x + 5 \cos y + 7 \sin z = 14$  then  
 $7 \tan \frac{x}{2} + 4 \cos y - 6 \cos z =$   
 1) 4      2) -3      3) 11      4) 5
27. If  $\sin \theta$  and  $\cos \theta$  are the roots of  
 $px^2 + qx + r = 0$  then  $q^2 - p^2 =$   
 1) 0      2)  $-2pr$       3)  $2qr$       4)  $2rp$
28. If  $0 \leq x \leq \pi$ ,  $4^{\sin^2 x} + 4^{\cos^2 x} = 5$ , then  $x =$   
 1)  $\frac{\pi}{6}$       2)  $\frac{\pi}{3}$       3)  $\frac{\pi}{4}$       4)  $\frac{\pi}{2}$

**LEVEL-II(H.W)-KEY**

- 01) 2    02) 4    03) 3    04) 1    05) 2    06) 3  
 07) 2    08) 4    09) 2    10) 1    11) 1    12) 2  
 13) 3    14) 3    15) 2    16) 3    17) 2    18) 2  
 19) 4    20) 1    21) 1    22) 3    23) 3    24) 1  
 25) 3    26) 3    27) 4    28) 4

**LEVEL-II(H.W)-HINTS**

1. lies in fourth quadrant substitute the corresponding values.
2.  $\sin \frac{3\pi}{2} = -1, \sin \frac{5\pi}{2} = 1$
- 3..  $\sin^2 A = \cos A \cdot \cot A \Rightarrow \sin^3 A = \cos^2 A$   
 $\Rightarrow \tan^2 A = \sec A$  and simplify
4.  $A + C = 180^\circ, B + D = 180^\circ$   
 Given  $\tan A = \frac{5}{12} > 0 \therefore 0 < A < \frac{\pi}{2} \Rightarrow \frac{\pi}{2} < C < \pi$   
 $\cos B = \frac{-3}{5} < 0 \therefore \frac{\pi}{2} < B < \pi \Rightarrow 0 < D < \frac{\pi}{2}$   
 $\therefore \cos C = \frac{-12}{13}; \tan D = \frac{4}{3}$

5.  $1 > 1^0 \Rightarrow \cos 1 < \cos 1^0$
6.  $2 = 2^c = 114^0$  (approx)
7. Use  $(\sec^2 \theta - \tan^2 \theta) = 1$
8.  $\sqrt{\cos^2 A} = \cos A$  then  $A \in Q_1, Q_4$
9. Eliminate  $\theta$
10. Use formula  $S_\infty = \frac{a}{1-r}$  for G.P
11.  $A + B = 180^\circ \Rightarrow \sin^2 A = \sin^2 B$   
 and  $A + B = 90^\circ \Rightarrow \sin^2 A + \sin^2 B = 1$
12.  $\frac{(x^2 + y^2)^p}{(xy)^q} = \frac{(a^2 \sin^4 \theta \cos^4 \theta)^p}{(a^2 \sin^5 \theta \cos^5 \theta)^q}$   
 $= \frac{a^{2p} (\sin \theta \cos \theta)^{4p}}{a^{2q} (\sin \theta \cos \theta)^{5q}}$   
 which is independent of  $\theta$ , if  $4p=5q$
13. Put  $\theta = 45^\circ$  and verify the options
14. squaring and adding
15.  $bc + \frac{1}{ck} + \frac{ak}{1+bk}$   
 $= \frac{\sin x}{k^2} + \frac{\cos x(1 + \cos x) + \sin^2 x}{\sin x(1 + \cos x)}$   
 $= \frac{a}{k} + \frac{1}{\sin x} = \frac{a}{k} + \frac{1}{ak} = \frac{1}{k} \left( a + \frac{1}{a} \right)$
16. divided with  $\sin^2 A$   
 we get  $\cot^2 A = \frac{p-n}{m-p}$
17. Put  $\sin x = b$  and  $\cos x = a$   
 $\therefore a^2 + b^2 = \cos^2 x + \sin^2 x = 1$
18. Use  $(a^4 + b^4) = (a^2 + b^2)^2 - 2a^2b^2$
19. In LHS divide the Num and Den by  $\cos \theta$ .
20.  $2 \tan \theta = (m+n); \quad 2 \sin \theta = (m-n)$   
 Eliminate  $\theta$  and use  $(\tan^2 \theta - \sin^2 \theta) = \tan^2 \theta \cdot \sin^2 \theta$
21. Put  $\theta = 0$
22. Range of  $\tan \theta$  is R

23. Use  $21\theta = \pi$

24. put  $n = 0$

25.  $(e^{\log 2})^{\frac{1}{1-\sin^2 x}} = 16 \Rightarrow 2^{\frac{1}{\cos^2 x}} = 2^4$

$\Rightarrow \sec^2 x = 4 \Rightarrow \tan^2 x = 3$

26. put  $x = \frac{\pi}{2}, y = 0, z = \frac{\pi}{2}$ .

27.  $\sin \theta + \cos \theta = \frac{-q}{p}, \sin \theta \cos \theta = \frac{r}{p}$

now  $(\sin \theta + \cos \theta)^2 = 1 + 2 \sin \theta \cos \theta$

28.  $x = 90^\circ$ , verify

**LEVEL - III**

1.  $\tan^2 \alpha = 1 - p^2$ , then  $\sec \alpha + \tan^3 \alpha \operatorname{cosec} \alpha =$

1)  $(2+p^2)^{\frac{3}{2}}$  2)  $(1+p^2)^{\frac{3}{2}}$

3)  $(2-p^2)^{\frac{3}{2}}$  4)  $(1-p^2)^{\frac{3}{2}}$

2.  $\frac{\pi}{2} < \alpha < \pi \Rightarrow \sqrt{\frac{1-\cos \alpha}{1+\cos \alpha}} + \sqrt{\frac{1+\cos \alpha}{1-\cos \alpha}} =$

1)  $2 \sec \alpha$  2)  $-2 \sec \alpha$

3)  $2 \operatorname{cosec} \alpha$  4)  $-2 \operatorname{cosec} \alpha$

3. If  $\sin \beta$  is the G.M. between  $\sin \alpha$  and  $\cos \alpha$

then  $(\cos \alpha - \sin \alpha)^2 - 2 \cos^2 \beta =$

1) 0 2) 1 3) 2 4) -1

4. If  $\sin^2 \theta = \frac{x^2 + y^2 + 1}{2x}$  then  $x$  must be

1) -3 2) -2 3) 1 4) 3

5.  $\sin x + \sin^2 x + \sin^3 x = 1 \Rightarrow \cos^6 x - 4 \cos^4 x + 8 \cos^2 x =$

1) 4 2) 2 3) 1 4) 0

6.  $\cos \theta + \cos^2 \theta = 1,$

$a \sin^{12} \theta + b \sin^{10} \theta + c \sin^8 \theta$

$+ d \sin^6 \theta = 1 \Rightarrow \frac{b+c}{a+d} =$

1) 2 2) 3 3) 4 4) 6

7.  $a \sin x = b \cos x = \frac{2c \tan x}{1 - \tan^2 x}$  and

$(a^2 - b^2)^2 = kc^2(a^2 + b^2) \Rightarrow k =$

1) 1 2) 2 3) 3 4) 4

8. The greatest among  $(\sin 1 + \cos 1),$

$(\sqrt{\sin 1} + \sqrt{\cos 1}), (\sin 1 - \cos 1)$  and 1 is

1)  $\sin 1 + \cos 1$  2)  $\sqrt{\sin 1} + \sqrt{\cos 1}$

3)  $\sin 1 - \cos 1$  4) 1

9. If  $a \sin^2 x + b \cos^2 x = c, b \sin^2 y + a \cos^2 y = d$

and  $a \tan x = b \tan y$  then  $\frac{a^2}{b^2} =$

1)  $\frac{(a-d)(c-a)}{(b-c)(d-b)}$  2)  $\frac{(b-c)(b-d)}{(a-c)(a-d)}$

3)  $\frac{(b-c)(d-b)}{(a-d)(c-a)}$  4)  $\frac{(d-a)(c-a)}{(b-c)(d-b)}$

10. If  $\tan^2 \alpha \tan^2 \beta + \tan^2 \beta \tan^2 \gamma$

$+ \tan^2 \gamma \tan^2 \alpha + 2 \tan^2 \alpha \tan^2 \beta \tan^2 \gamma = 1$  then

the value of  $\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma =$

1) 0 2) -1 3) 1 4) 2

11. Let  $\theta \in \left(0, \frac{\pi}{4}\right)$  and  $t_1 = (\tan \theta)^{\tan \theta},$

$t_2 = (\tan \theta)^{\cot \theta}, t_3 = (\cot \theta)^{\tan \theta},$

$t_4 = (\cot \theta)^{\cot \theta}$  then

1)  $t_1 > t_2 > t_3 > t_4$  2)  $t_4 > t_3 > t_1 > t_2$

3)  $t_3 > t_1 > t_2 > t_4$  4)  $t_2 > t_3 > t_1 > t_4$

12. Two arcs of same length of two different circles subtended angles of  $25^\circ$  and  $75^\circ$  at their centres respectively. Then the ratio of the radii of the circles is

1) 3 : 1 2) 1 : 3 3) 1 : 2 4) 2 : 1

13. If  $f_k(x) = \frac{1}{k}(\sin^k x + \cos^k x)$  where

$x \in R, k \geq 1$  then  $f_4(x) - f_6(x) =$

(MAINS-2014)

1)  $\frac{1}{6}$  2)  $\frac{1}{3}$  3)  $\frac{1}{9}$  4)  $\frac{1}{12}$

14. The expression  $\frac{\tan A}{1 - \cot A} + \frac{\cot A}{1 - \tan A}$  can be written as (MAINS-2013)

- 1)  $\sin A \cos A + 1$       2)  $\sec A \cos ec A + 1$   
3)  $\tan A + \cot A$       4)  $\sec A + \cos ec A$

15. In a  $\Delta PQR$ , if  $3 \sin P + 4 \cos Q = 6$  and  $4 \sin Q + 3 \cos P = 1$  then the acute angle 'R' is equal to [AIEEE-2012]

- 1)  $\frac{\pi}{4}$       2)  $\frac{3\pi}{4}$       3)  $\frac{5\pi}{6}$       4)  $\frac{\pi}{6}$

LEVEL-III- KEY

- 01) 3      02) 3      03) 4      04) 3      05) 1      06) 2  
07) 4      08) 2      09) 1      10) 3      11) 2      12) 1  
13) 4      14) 2      15) 4

LEVEL-III- HINTS

- $\sec \alpha \left[ 1 + \frac{\sin^3 \alpha \cos \alpha}{\cos^3 \alpha \sin \alpha} \right]$   
 $= \sec \alpha [1 + \tan^2 \alpha] = \sec^3 \alpha$   
 $= (\sec^2 \alpha)^{3/2} = (2 - P^2)^{3/2}$
- Simplifying we get  $\frac{2}{\sqrt{\sin^2 \alpha}} = 2 \cos ec \alpha$   
 $\therefore \sin \alpha$  is positive in II quadrant.
- $\sin^2 \beta = \sin \alpha \cos \alpha$
- $\sin^2 \theta \leq 1$
- $\sin x (1 + \sin^2 x) = \cos^2 x$   
Squaring both sides we get  
 $(1 - \cos^2 x)(2 - \cos^2 x)^2 = \cos^4 x$   
Simplify further
- $\cos \theta = 1 - \cos^2 \theta \Rightarrow \cos \theta = \sin^2 \theta$   
 $\Rightarrow \cos^2 \theta = \sin^4 \theta \Rightarrow (\sin^4 \theta + \sin^2 \theta) = 1$   
Cubing both sides we get the answer
- Take  $a \sin x = b \cos x = c \tan 2x = l$   
 $\Rightarrow \sin x = \frac{l}{a}, \cos x = \frac{l}{b}, \tan 2x = \frac{l}{c}$   
Elimination  $l$  we get  
 $(a^2 - b^2)^2 = 4c^2 (a^2 + b^2) \Rightarrow K = 4$

- $\sqrt{\sin 1} > \sin 1 > \sin^2 1$   
 $\sqrt{\cos 1} > \cos 1 > \cos^2 1$   
Adding  $(\sqrt{\sin 1} + \sqrt{\cos 1}) > (\sin 1 + \cos 1)$   
 $> (\sin^2 1 + \cos^2 1)$   
 $\Rightarrow (\sqrt{\sin 1} + \sqrt{\cos 1}) > (\sin 1 + \cos 1) > 1$
- $a \sin^2 x + b \cos^2 x = c$   
 $\Rightarrow a \tan^2 x + b = c \sec^2 x$   
 $\Rightarrow (a - c) \tan^2 x = c - b \Rightarrow \tan^2 x = \frac{c - b}{a - c}$   
similarly  $\tan^2 y = \frac{d - a}{b - d}$   
but  $a \tan x = b \tan y \Rightarrow \frac{a^2}{b^2} = \frac{\tan^2 y}{\tan^2 x}$
- Let  $\tan^2 \alpha = x, \tan^2 \beta = y, \tan^2 \gamma = z$   
 $\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma = \sum \frac{\tan^2 \alpha}{1 + \tan^2 \alpha} = \sum \frac{x}{1 + x}$   
 $= \frac{(x + y + z) + (xy + yz + zx + 2xyz) + xy + yz + zx + xyz}{(1 + x)(1 + y)(1 + z)}$   
 $= \frac{1 + x + y + z + xy + yz + zx + xyz}{(1 + x)(1 + y)(1 + z)} = 1$
- In  $\left(0, \frac{\pi}{4}\right), \tan \theta < 1$  and  $\cot \theta > 1$   
 $(\cot \theta)^{\cot \theta} > (\cot \theta)^{\tan \theta} > (\tan \theta)^{\tan \theta} > (\tan \theta)^{\cot \theta}$   
 $\therefore t_4 > t_3 > t_1 > t_2$
- Length of arc,  $l = r\theta$   
 $\therefore \frac{r_1}{r_2} = \frac{l/\theta_1}{l/\theta_2} = \frac{\theta_2}{\theta_1} = \frac{75}{25} = \frac{3}{1}$
- $f_4(x) = 1 - 2 \sin^2 \theta \cos^2 \theta$  and  
 $f_6(x) = 1 - 3 \sin^2 \theta \cos^2 \theta$   
 $\frac{\sin A}{\cos A} + \frac{\cos A}{\sin A} = \frac{\sin^2 A}{\cos A(\sin A - \cos A)} + \frac{\cos^2 A}{\sin A(\cos A - \sin A)}$   
 $= \frac{\sin^3 A - \cos^3 A}{\sin A \cos A(\sin A - \cos A)} = \frac{\sin^2 A + \sin A \cos A + \cos^2 A}{\sin A \cos A}$   
 $= \frac{1 + \sin A \cos A}{\sin A \cos A} = \sec A \cos ec A + 1$
- Squaring both the equations and adding





## LEVEL - IV

1. **Assertion A:** In a right angled triangle

$$\sin^2 A + \sin^2 B + \sec^2 C = 2.$$

**Reason R:** If  $\alpha, \beta$  are complementary angles

$$\text{then } \sin^2 \alpha + \sin^2 \beta = 1$$

- 1) A, R are true and R is the correct explanation of A  
 2) A, R are true and R is not the correct explanation of A  
 3) A is true, R is false  
 4) A is false, R is true
2. **Observe the following lists. Let**

$$\frac{\sin x}{a} = \frac{\cos x}{b} = \frac{\tan x}{c} = k.$$

**List - I**

**List-II**

1)  $bc$

a)  $\frac{1}{b^2 k^4}$

2)  $a^2 + b^2$

b)  $\frac{1}{ak}$

3)  $\frac{1}{ck} + \frac{ak}{1+bk}$

c)  $\frac{a}{k}$

4)  $a^2 + b^2 + c^2$

d)  $\frac{1}{k^2}$

The correct match for List-I from List-II is

- 1) 1 - c, 2 - d, 3 - b, 4 - a  
 2) 1 - d, 2 - a, 3 - c, 4 - b  
 3) 1 - a, 2 - b, 3 - d, 4 - c  
 4) 1 - b, 2 - c, 3 - a, 4 - d

3. **Observe the following lists.**

**List-I**

**List-II**

I.  $\sin^2 \frac{2\pi}{3} + \cos^2 \frac{5\pi}{6} - \tan^2 \frac{3\pi}{4}$

a)  $\frac{15}{2}$

II.  $\sin^2 \frac{11\pi}{6} + \cos^2 \frac{7\pi}{6} - \tan^2 \frac{5\pi}{4}$

b)  $-\frac{1}{2}$

III.  $\sin^2 \frac{3\pi}{4} + \sec^2 \frac{5\pi}{3} + \tan^2 \frac{2\pi}{3}$

c) 0

IV.  $\cos^2 \frac{2\pi}{3} + \sin^2 \frac{5\pi}{3} - \frac{3}{2} \tan^2 \frac{3\pi}{4}$

d)  $1/2$

The correct match for List-I from List-II is

- 1) I-d, II-c, III-a, IV-b 2) I-a, II-b, III-c, IV-d  
 3) I-b, II-a, III-d, IV-c 4) I-c, II-d, III-b, IV-a

4. **Statement I:** The number of values of  $\theta$  satisfying  $\sec \theta + \cos \theta = 1$  is 2.

**Statement II:** If  $7 \sin^2 \alpha + 3 \cos^2 \alpha = 4$ , then

$$\tan \alpha = \sqrt{3}.$$

Which of the above statements is correct?

- 1) only I is true 2) only II is true  
 3) Both I and II are true  
 4) Neither I nor II is true
5. If  $\alpha = \cos 10^\circ - \sin 10^\circ$ ,  
 $\beta = \cos 45^\circ - \sin 45^\circ$ ,  
 $\gamma = \cos 70^\circ - \sin 70^\circ$  then the descending order of  $\alpha, \beta, \gamma$  is
- 1)  $\alpha, \beta, \gamma$  2)  $\gamma, \beta, \alpha$  3)  $\alpha, \gamma, \beta$  4)  $\beta, \alpha, \gamma$

6. **Match the following:**

**List - I**

**List - II**

1)  $3 \tan x + 27 \cot x \geq$  a) 24  
 $(x \in Q_1)$

2)  $5 \sec^2 x + 125 \cos^2 x \geq$  b) 18

3)  $16 \operatorname{cosec}^2 x + 9 \sin^2 x \geq$  c) 50

1) 1-a, 2-b, 3-c 2) 1-c, 2-a, 3-b

3) 1-b, 2-c, 3-a 4) 1-c, 2-b, 3-a

7. **A = tan1, B = tan2, C = tan3, then the descending order of A, B, C is**

1) A, B, C 2) C, B, A 3) A, C, B 4) B, C, A

## LEVEL-IV-KEY

01) 4 02) 1 03) 1 04) 4 05) 1 06) 3 07) 3

## LEVEL-IV-HINTS

1. In a right angled triangle  
 $\sin^2 A + \sin^2 B + \sec^2 C \neq 2$
2. Simplify
3. Simplify
4. I.  $(\sec \theta + \cos \theta) = 1$  is not possible for  $\theta \in R$   
 II. Divide given equation by  $\cos^2 \alpha$
5.  $\alpha$  is positive,  $\beta$  is zero,  
 $\gamma$  is negative
6.  $A.M \geq G.M$   
 $a \tan x + b \cot x \geq 2\sqrt{ab}$
7.  $1 = 57^\circ$   
 $2 = 114^\circ$   
 $3 = 171^\circ$  (Aprox)